

31-08-2023



1

Chemical Reaction Engineering objectives

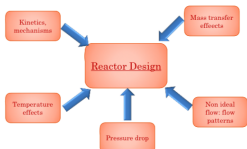
- Provide an overview of **chemical kinetics** and **reactor design** at basic to intermediate level
- Apply the concepts of reaction rate, then stoichiometry and equilibrium to the analysis of different chemicals reacting systems.
 - derivation of rate expression from the reaction mechanism
 - equilibrium or steady state assumptions for certain reactions.
- Design of Chemical reactors via synthesis of chemical kinetics and mass and energy balances.

The broad goal of this course is to provide students with theoretical and analytical background so that they can understand the chemical kinetics and the reactor design to tackle the sort of complex problem.

BITS Pilani, Pilani Campus

2

Objectives



4

Introduction to KRD

Why we need KRD?

- KRD is the only subjects which distinguish between the chemical engineers to the others
- Reactor is the central features for most of the chemical processes, and even if the separation cost dominates the reactor often determines the separation cost.
- Chemical reactions are ubiquitous in nature and industry
 - mathematical description of chemical reactions and chemical reactors.
 - allow us to predict and understand the different reactions and reactors.

BITS Pilani, Pilani Campus

6

Chemist vs Chemical Engineer



BTS Plant, Plant Campus

7



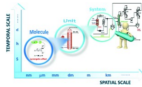
It cannot be done in this way !

BTS Plant, Plant Campus

8

Chemical Reaction Engineering – New frontiers

- New analytical instrumentation providing quantitative information on species and mechanisms
- New computational resources – hardware and software
- Multi-scale approach



BTS Plant, Plant Campus

9

Research in catalysis involve a multi-discipline approach

- Reaction kinetics and mechanism
 - > Reaction paths, intermediate formation & action, interpretation of results obtained under various conditions, generalising reaction types & schemes, predict catalyst performance...
- Catalyst development
 - > Material synthesis, structure properties, catalyst stability, compatibility...
- Analysis techniques
 - > Detection limits in terms of dimension of time & size and under extreme conditions (T, P) and accuracy of measurements, microscopic techniques, sample preparation techniques...
- Reaction modelling
 - > Elementary reactions and rates, quantum mechanical/chemistry, physical chemistry...
- Reactor modelling
 - > Mathematical interpretation and representation, the numerical method, micro-kinetics, structure and efficiency of heat and mass transfer in relation to reactor design...
- Catalytic process
 - > Heat and mass transfers, energy balance and efficiency of process...

BTS Plant, Plant Campus

10

Softwares

- **Molecular dynamics (MD)**: Studying the physical movements of atoms and molecules
 - Molecular studio
 - Jmol/ LAMMPS
- **Density functional theory (DFT)** is a computational quantum mechanical modelling method used to investigate the electronic structure
 - NWChem, OpenMX
- **CHEMKIN, DETCHEM, GASEQ, etc ...**

11

Let's start KRD

- **Chemical Reaction Engineering (CRE)** is the field that studies the **rates** and **mechanisms** of chemical reactions and the **design of the reactors** in which they take place.
- Chemical reaction engineering, as that engineering activity which is concerned with exploitation of **chemical reactions** on a **commercial scale**

12

Chapter 1: Objectives

- Rate of reaction (r_f)
- Extent of reaction, conversion
- General Mole Balance Equations
 - **Batch Reactor**
 - **Continuously Stirred Tank Reactor**
 - **Plug flow Reactor**
 - **Packed Bed Reactor**

13

Chemical Process in general



- ✓ The chemical reactor must be regarded as the **heart of a chemical process**. The reactor is the equipment in which conversion of feedstock to desired product takes place.

14

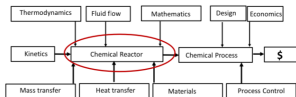
What do we need to know?

- The chemical engineer almost never encounters
 - a single reaction in an ideal single phase
 - isothermal reactor, ideal reactors
- Real reactors are extremely complex with multiple reactions, multiple phases, and intricate flow patterns within the reactor and in inlet and outlet streams

BTS Ploem, Ploem Campus

Information needed?

- Analysis of chemical reactors incorporates essentially all the material in the chemical engineering curriculum



BTS Ploem, Ploem Campus

16

20

Sources?

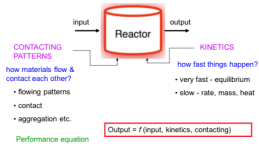
- Chemicals that can be sold for **high prices** from inexpensive raw materials. This involves finding a chemical reactor system that will do this better than the competition, finding cheap and abundant raw materials, finding a **good and reliable** market for the product, and disposing of byproducts
- Chemical and Engineering News- weekly- ACS*
- Chemical Marketing Report*
- Chemical weekly*

BTS Ploem, Ploem Campus

21

Reactor performance

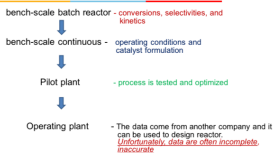
Information needed to predict the reactor behavior:



BTS Ploem, Ploem Campus

22

Source of the essential data for reactor design?



BTS Plant, Plant Campus

25

Chemical Engineer Tasks?

- Maintain and operate a process
- Fix some perceived problem or
- Increase capacity or selectivity at minimum cost



BTS Plant, Plant Campus

26

Representation of reaction

- Consider the species in a reaction: $A_1, A_2, A_3, \dots, A_i, S$ is number of species

General representations is

$$\sum_{j=1}^S \nu_j A_j = 0$$

The linear independent reactions in a reaction network is equivalent to the linear independent rows in the corresponding stoichiometric matrix.



BTS Plant, Plant Campus

28

Progress of a Reaction

- Consider a reaction $\sum \nu_j A_j = 0$ taking place in a closed system

n_{j0} = number of moles of species j present initially
 n_j = number of moles of species j at any time t

- Molar extent of reaction - α

$$\alpha = \frac{n_j - n_{j0}}{\nu_j}$$

- Properties of α

↳ defined for the reaction

$$\alpha = \frac{n_j - n_{j0}}{\nu_j} = \frac{n_s - n_{s0}}{\nu_s}$$

$$\Rightarrow \alpha_j = n_{j0} + \frac{\nu_j}{\nu_s} (n_s - n_{s0})$$

↳ Extensive property in moles

↳ Always positive



BTS Plant, Plant Campus

29

Conversion

- Conversion X

$$X = \frac{n_{j0} - n_j}{n_{j0}}$$

- Stoichiometrically limiting species "k"

$$\min \left(-\frac{n_{k0}}{v_j} \right)$$

Reactions

- A chemical species is said to have reacted when it has lost its chemical identity.
- There are three ways for a species to lose its identity:

- Decomposition** $\text{CH}_3\text{CH}_3 \rightarrow \text{H}_2 + \text{H}_2\text{C}=\text{CH}_2$
- Combination** $\text{N}_2 + \text{O}_2 \rightarrow 2 \text{NO}$
- Isomerization** $\text{C}_2\text{H}_5\text{CH}=\text{CH}_2 \rightarrow \text{CH}_2=\text{C}(\text{CH}_3)_2$

30

31

Reactions

- Consider a reaction $\sum \nu_j A_j = 0$, taking place in a **closed, isothermal, constant pressure system**
- Reaction rate: Speed at which a chemical reaction proceeds, in terms of amount of product formed or amount of reactant consumed per unit time

$$r = \frac{1}{V} \frac{d\alpha}{dt} \quad \alpha = \frac{n_j - n_{j0}}{V_j}$$

$$r_j = \frac{1}{V} \frac{dn_j}{dt}$$

$$\begin{aligned} r &= r(T, P, y_1, y_2, \dots, y_{N-1}) \\ &= r(T, P, C_1, C_2, \dots, C_{N-1}) \\ &= r(T, C_1, C_2, \dots, C_{N-1}, C_N) \end{aligned}$$

32

Variables affecting the rate of reaction

Homogeneous reaction

- Temperature
- Pressure
- Composition



Heterogeneous reaction

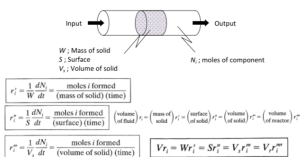
- Temperature
- Pressure
- Composition

- Mass transfer
- Heat transfer



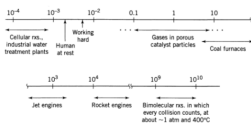
33

Heterogeneous Reactions



34

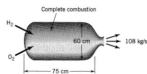
Reaction Rates



35

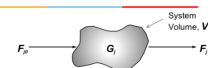
Example

- A rocket engine, shown in Fig., burns a stoichiometric mixture of fuel (liquid hydrogen) in oxidant (liquid oxygen). The combustion chamber is cylindrical, 75 cm long and 60 cm in diameter, and the combustion process produces 108 kg/s of exhaust gases. If combustion is complete, find the rate of reaction of hydrogen and of oxygen.



36

General mole balance



$$\left[\begin{array}{c} \text{Molar Flow} \\ \text{Rate of} \\ \text{Species } j \text{ in} \end{array} \right] - \left[\begin{array}{c} \text{Molar Flow} \\ \text{Rate of} \\ \text{Species } j \text{ out} \end{array} \right] + \left[\begin{array}{c} \text{Molar Rate} \\ \text{Generation} \\ \text{of Species } j \end{array} \right] = \left[\begin{array}{c} \text{Molar Rate} \\ \text{Accumulation} \\ \text{of Species } j \end{array} \right]$$

$$F_{in} - F_j + G_j = \frac{dN_j}{dt}$$

$$\left(\frac{\text{mole}}{\text{time}} \right) - \left(\frac{\text{mole}}{\text{time}} \right) + \left(\frac{\text{mole}}{\text{time}} \right) = \left(\frac{\text{mole}}{\text{time}} \right)$$

37

General mole balance contd...

- If spatially uniform:

$$G_j = r_j V$$

- If NOT spatially uniform:



38

General mole balance contd...

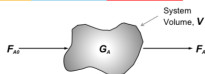
$$G_j = \sum_{i=1}^n r_{ji} \Delta V_i$$

Take limit

$$G_j = \sum_{i=1}^n r_{ji} \Delta V_i \xrightarrow[n \rightarrow \infty]{i \rightarrow 1} \lim \Delta V \rightarrow 0} = \int r_j dV$$

39

General mole balance contd...



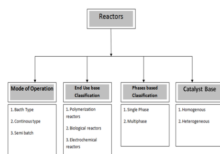
General Mole Balance on System Volume V

$$\text{In} - \text{Out} + \text{Generation} = \text{Accumulation}$$

$$F_{A0} - F_A + \int r_A dV = \frac{dN_A}{dt}$$

40

Ideal Reactors



41

Batch Reactor

- Uniform composition, everywhere in the reactor, but the composition changes with time.
- A "batch" of reactants is introduced into the reactor operated at the desired conditions until the target conversion is reached.
- Batch reactors are typically tanks in which stirring of the reactants is achieved using internal impellers, gas bubbles, or a pump-around loop where a fraction of the reactants is removed and externally recirculated back to the reactor.



BTS Plant, Plant Campus

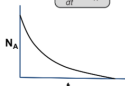
42

Batch Reactor- Mole balance

Well-Mixed $\int r_A dV = r_A V$

$$\frac{dN_A}{dt} = r_A V$$

$$t = \int_{N_{A0}}^{N_A} \frac{dN_A}{-r_A V}$$



Time necessary to reduce the number of moles of A from N_{A0} to N_A .

BTS Plant, Plant Campus

43

Ideal Continuous Flow Reactor

- Reactants are added and products removed continuously at a constant mass flow rate. Large daily production rates are mostly conducted in continuous equipment.

- Continuously Stirred Tank Reactor (CSTR)
- Plug Flow Reactor (PFR)
- Packed Bed Reactor (PBR)

BTS Plant, Plant Campus

44

Ideal CSTR

- Uniformly mixed, same composition every where with in the reactor and at the exit
- Primarily in homogeneous liquid-phase flow reactions

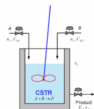
Advantages:

- Good temperature control
- Easy to construct

Disadvantage:

- Conversion is low

<https://www.youtube.com/watch?v=J5RqQ8pDDeU>



BTS Plant, Plant Campus

45

Ideal CSTR- Mole balance

$$\text{In} - \text{Out} + \text{Generation} = \text{Accumulation}$$

$$F_{A0} - F_A + \int r_A dV = \frac{dN_A}{dt}$$

$$V = \frac{F_{A0} - F_A}{-r_A}$$



CSTR volume necessary to reduce the molar flow rate from F_{A0} to F_A .

BITS Pilani, Pilani Campus

46

Ideal PFR

- It is characterized by the fact that the flow of fluid through the reactor is orderly with no element of fluid overtaking or mixing with any other element ahead or behind.
- May be lateral mixing; however, there must be no mixing or diffusion along the flow path

<https://youtu.be/wjNHuZiHTgA>



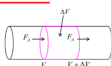
BITS Pilani, Pilani Campus

47

Ideal PFR- mole balance

$$\left[\text{In} \right]_{at V} - \left[\text{Out} \right]_{at V + \Delta V} + \left[\text{Generation} \right]_{in \Delta V} = 0$$

$$F_A|_V - F_A|_{V+\Delta V} + r_A \Delta V = 0$$



$$\frac{dF_A}{dV} = r_A$$

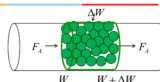
The integral form is: $V = \int_{F_{A0}}^{F_A} \frac{dF_A}{r_A}$

This is the volume necessary to reduce the entering molar flow rate (mol/s) from F_{A0} to the exit molar flow rate of F_A .

BITS Pilani, Pilani Campus

48

Ideal PBR- mole balance



$$F_A(W) - F_A(W + \Delta W) + r'_A \Delta W = \frac{dN_A}{dt}$$

Steady State $\frac{dN_A}{dt} = 0$

$$\lim_{\Delta W \rightarrow 0} \frac{F_A|_{W+\Delta W} - F_A|_W}{\Delta W} = r'_A$$

BITS Pilani, Pilani Campus

49

31-08-2023

Ideal PBR- mole balance

Rearrange:

$$\frac{dF_A}{dW} = r'_A$$

The integral form to find the catalyst weight is:

$$W = \int_{F_{A0}}^{F_A} \frac{dF_A}{r'_A}$$

PBR catalyst weight necessary to reduce the entering molar flow rate F_{A0} to molar flow rate F_A .

BTS Plant, Plant Campus

50

Ideal Reactor Mole Balances Summary

Reactor	Differential	Algebraic	Integral
Batch	$\frac{dN_A}{dt} = r_A V$		$N_A = \int_{F_{A0}}^{F_A} \frac{dF_A}{r'_A}$ 
CSTR		$r_A = \frac{F_{A0} - F_A}{V}$	Can you guess? 
PBR	$\frac{dF_A}{dW} = r'_A$		$F_A = \int_{F_{A0}}^{F_A} \frac{dF_A}{r'_A}$ 
PFR	$\frac{dF_A}{dV} = r'_A$		$F_A = \int_{F_{A0}}^{F_A} \frac{dF_A}{r'_A}$ 

BTS Plant, Plant Campus

51

Ideal Reactors

Batch



Uniform composition everywhere in the reactor, but of course the composition changes with time.

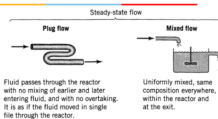
$$\frac{dN_A}{dt} = r_A V$$

$$t = \int_{N_{A0}}^{N_A} \frac{dN_A}{-r_A V}$$

BTS Plant, Plant Campus

52

Ideal Reactors



$$\frac{dF_A}{dV} = r'_A$$

$$V = \int_{F_{A0}}^{F_A} \frac{dF_A}{r'_A}$$

$$V = \frac{F_{A0} - F_A}{-r'_A}$$

BTS Plant, Plant Campus

53



Problem



- The reaction $A \rightarrow B$ is to be carried out isothermally in a continuous-flow reactor. Calculate both the CSTR and PFR reactor volumes necessary to consume 99% of A ($C_p = 0.01C_{A0}$), when the entering molar flow rate is 5 mol/h, assuming the reaction rate $-r_A$ is:

a. $-r_A = k$ with $k = 0.05 \text{ mol/h} \cdot \text{dm}^3$

b. $-r_A = kC_A$ with $k = 0.0001 \text{ s}^{-1}$

c. $-r_A = kC_A^2$ with $k = 3 \text{ dm}^3/\text{mol} \cdot \text{hr}$

The entering volumetric flow rate is $10 \text{ dm}^3/\text{h}$

- d. If the above reaction is carried out in a batch reactor. Calculate the batch to get 99% conversion of A in a, b & c.